

OOPS ASSIGNMENT

Question1.

Code :

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
int main() {
```

```
    ofstream outFile("NUM.TXT");
```

```
    if (!outFile) {
```

```
        cout << "Error opening file!" << endl;
```

```
        return 1;
```

```
    }
```

```
    for (int i = 1; i <= 200; i++) {
```

```
        outFile << i << endl;
```

```
    }
```

```
    outFile.close();
```

```
    cout << "Numbers 1 to 200 written to NUM.TXT successfully!" << endl;
```

```
    return 0;
```

```
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Numbers 1 to 200 written to NUM.TXT successfully!
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question2.

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <cctype>
```

```
using namespace std;
```

```
void countAlphabets() {
    ifstream inFile("NOTES.TXT");
    if (!inFile) {
        cout << "Error opening file!" << endl;
        return;
    }
    char ch;
    int count = 0;
    while (inFile.get(ch)) {
        if (isalpha(ch)) {
            count++;
        }
    }
    inFile.close();
    cout << "Number of alphabets in NOTES.TXT: " << count << endl;
}

int main() {
    countAlphabets();
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Error opening file!
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question3.

Code:

```
#include <iostream>

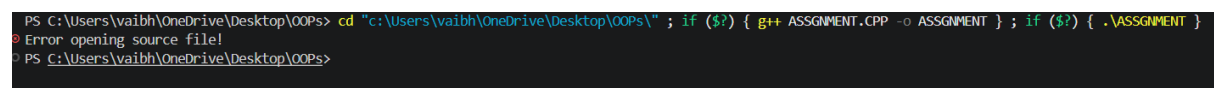
#include <fstream>

using namespace std;

int main() {
```

```
ifstream sourceFile("source.txt");
ofstream destFile("destination.txt");
if (!sourceFile) {
    cout << "Error opening source file!" << endl;
    return 1;
}
if (!destFile) {
    cout << "Error creating destination file!" << endl;
    return 1;
}
char ch;
while (sourceFile.get(ch)) {
    destFile.put(ch);
}
sourceFile.close();
destFile.close();
cout << "File copied successfully!" << endl;
return 0;
}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Error opening source file!
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question4.

Code:

```
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;
```

```
int main() {
    char str[100];
```

```
int length;

cout << "Enter a string: ";
cin.getline(str, 100);

length = strlen(str);
cout << "Length of string: " << length << endl;

ofstream outFile("string.txt");
if (!outFile) {
    cout << "Error opening file!" << endl;
    return 1;
}
outFile << str;
outFile.close();
cout << "String stored in file!" << endl;

ifstream inFile("string.txt");
if (!inFile) {
    cout << "Error opening file!" << endl;
    return 1;
}
char ch;
cout << "Content from file: ";
while (inFile.get(ch)) {
    cout << ch;
}
cout << endl;
inFile.close();
return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Enter a string: i am a student
Length of string: 14
String stored in file!
Content from file: i am a student
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> 
```

Question5.

Code:

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
int main() {
```

```
    // (a) Create file A-Z
```

```
    ofstream outFile("A-Z.txt");
```

```
    for (char ch = 'A'; ch <= 'Z'; ch++) {
```

```
        outFile.put(ch);
```

```
    }
```

```
    outFile.close();
```

```
    ifstream inFile("A-Z.txt");
```

```
    inFile.seekg(9);
```

```
    char ch;
```

```
    inFile.get(ch);
```

```
    cout << "10th character: " << ch << endl;
```

```
    inFile.close();
```

```
    // (b) Overwrite 5th position using seekp()
```

```
    ofstream outFile2("numbers.txt");
```

```
    for (int i = 1; i <= 10; i++) {
```

```
        outFile2 << i << " ";
```

```
    }
```

```
    outFile2.close();
```

```
fstream file("numbers.txt", ios::in | ios::out);  
file.seekp(0);  
file << "99";  
file.close();  
cout << "5th position overwritten with 99" << endl;
```

```
// (c) Find file size using tellg()  
ifstream file3("A-Z.txt");  
file3.seekg(0, ios::end);  
int size = file3.tellg();  
cout << "File size: " << size << " bytes" << endl;  
file3.close();
```

```
// (d) Read last character using seekg()  
ifstream file4("A-Z.txt");  
file4.seekg(-1, ios::end);  
file4.get(ch);  
cout << "Last character: " << ch << endl;  
file4.close();
```

```
// (e) Create data.txt with multiple lines  
ofstream dataFile("data.txt");  
dataFile << "Line 1: Hello World" << endl;  
dataFile << "Line 2: C++ Programming" << endl;  
dataFile << "Line 3: File Handling" << endl;  
dataFile.close();
```

```
ifstream readFile("data.txt");  
readFile.seekg(10);  
int pos = readFile.tellg();
```

```
cout << "Current position: " << pos << endl;

cout << "Remaining content:" << endl;

char c;

while (readFile.get(c)) {

    cout << c;

}

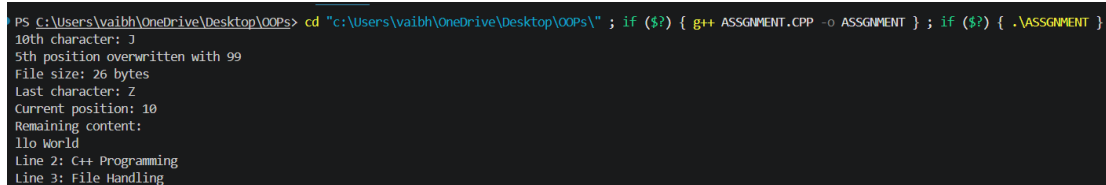
cout << endl;

readFile.close();

return 0;

}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
10th character: J
5th position overwritten with 99
File size: 26 bytes
Last character: Z
Current position: 10
Remaining content:
Hello World
Line 2: C++ Programming
Line 3: File Handling
```

Question6.

Code:

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
int main() {

    fstream file("hello.txt", ios::out | ios::in | ios::trunc);

    if (!file) {

        ofstream createFile("hello.txt");

        createFile.close();

        file.open("hello.txt", ios::out | ios::in);

    }

    string text = "Hello World";

    cout << "Writing characters:" << endl;
```

```
for (int i = 0; i < text.length(); i++) {  
    file.put(text[i]);  
    int pos = file.tellp();  
    cout << "Written " << text[i] << " at position: " << pos << endl;  
}  
  
file.seekp(6);  
file << "C++ ";  
  
file.seekg(0);  
char ch;  
cout << "\nUpdated content: ";  
while (file.get(ch)) {  
    cout << ch;  
}  
cout << endl;  
file.close();  
return 0;  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }  
Writing characters:  
Written 'H' at position: 1  
Written 'e' at position: 2  
Written 'l' at position: 3  
Written 'l' at position: 4  
Written 'o' at position: 5  
Written ' ' at position: 6  
Written 'W' at position: 7  
Written 'o' at position: 8  
Written 'r' at position: 9  
Written 'l' at position: 10  
Written 'd' at position: 11  
  
Updated content: Hello C++ d  
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```